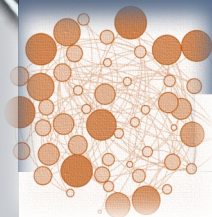
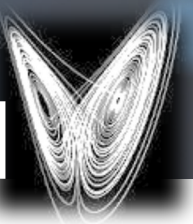


Lecture : Flocking of birds

<https://youtu.be/MIFtHuXPbv4>



Lab For Networks And Nonlinear Dynamics
(LN²D)



Engineering Systems



What Is The Agent?

- **Agent:** *system with two important capabilities:*
 - **Autonomy:** *capable of autonomous action* – of deciding for themselves what they need to do in order to satisfy their objectives ;
 - **Interactions:** *capable of interacting with other agents* - the kind of social activity that we all engage in every day of our lives: cooperation, competition, negotiation, and the like.
- **Examples:** *Individual, insect, bird, fish, people, robot, ...*

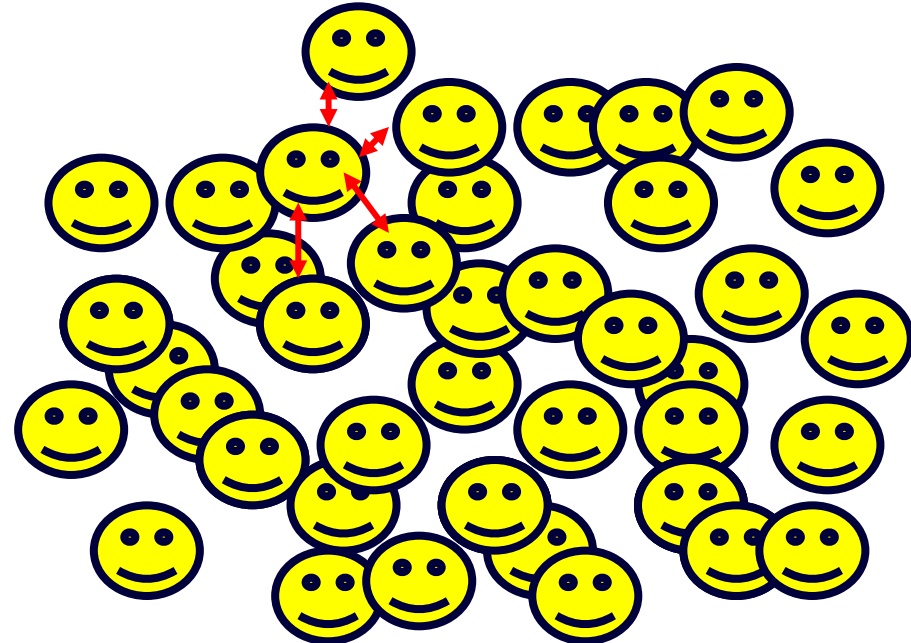
Multi-Agent System (MAS)

- **MAS**

- Many agents
- Local interactions between agents
- Collective behavior in the population level
 - More is different.---Philp Anderson, 1972
 - e.g., Phase transition, coordination, synchronization, consensus, clustering, aggregation,

- **Examples:**

- Physical systems
- Biological systems
- Social and economic systems
- Engineering systems



Biological Systems



Flocking of Birds



Ant Colony



Bee Colony



Bacteria Colony

Animal collective behaviour

Heterogeneous, "aggregated" patterns in the spatial distributions of individuals are almost universal across living organisms. What are the individual-level behavioral traits that give rise to these features? Mathematical models of social behaviors such as schooling in fishes, swarming in ants, of flocking in birds provide a promising avenue to address some of these questions.

Physical aggregation can be regarded as part of a continuum in group integration, Honeybees hives, cetacean pods and human communities are examples of these integrated groups.

Animal collective behaviour

Heterogeneous, "aggregated" patterns in the spatial distributions of individuals are almost universal across living organisms. What are the individual-level behavioral traits that give rise to these features? Mathematical models of social behaviors such as schooling in fishes, swarming in ants, or flocking in birds provide a promising avenue to address some of these questions.

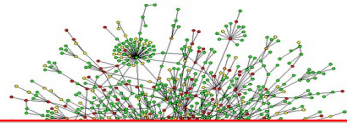
Physical aggregation can be regarded as part of a continuum in group integration, Honeybees hives, cetacean pods and human communities are examples of these integrated groups.

Between these extremes of group integration are what could be considered "prototypical" animal aggregation, herds, swarms, flocks and schools. Many of these groups are extremely large (e.g. a school of a billion herring). Individuals in such groups interact with a neighborhood of the other members, but those may represent a quite small fraction of the group as a whole. This suggests that mechanisms that maximize information transfer among individuals could be beneficial.

Dynamic patterns and movement are necessary characteristics of many biological aggregation, and are perhaps adaptive responses from environment phenomena. For example, fish schools display complex emergent properties such as coordinated motion and directed activity. Compression, vacuole, fountain, and flash expansion are all maneuvers that maximize predatory risk only if all members perform them correctly.

From Local Rules to Collective Behavior

**Phase transition, coordination, synchronization,
consensus, clustering, aggregation,**

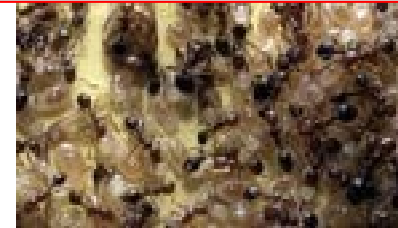


pattern

*A basic problem: How locally interacting agents
lead to the collective behavior of the overall
systems?*



Crowd Panic



swarm intelligence

Modeling of MAS

- **Distributed/Autonomous**
- **Local interactions/rules**
- **Neighbors may be dynamic**
- **May have no physical connections**

Assumption

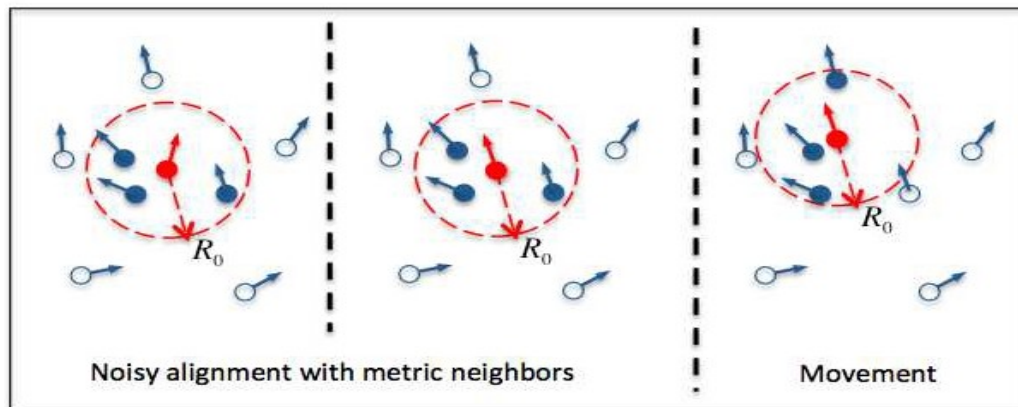
Each agent

- makes decision according to **local information**
- has the tendency **to behave as other agents do in its neighborhood.**

Vicsek Model (T. Vicsek *et al.*, PRL, 1995)

$$\vec{x}_i(t + \Delta t) = \vec{x}_i(t) + \vec{v}_i(t)\Delta t \quad (1)$$

A



- Velocities, having fixed magnitude v_0 , denoted by $\vec{v}_i \quad i = 1; \dots; N$.
- In two dimensions, the direction of motion of the i -th particle can be described via a single angle, θ_i , so $\vec{v}_i = (v_0 \cos \theta_i, v_0 \sin \theta_i)$

Vicsek Model (T. Vicsek *et al.*, PRL, 1995)

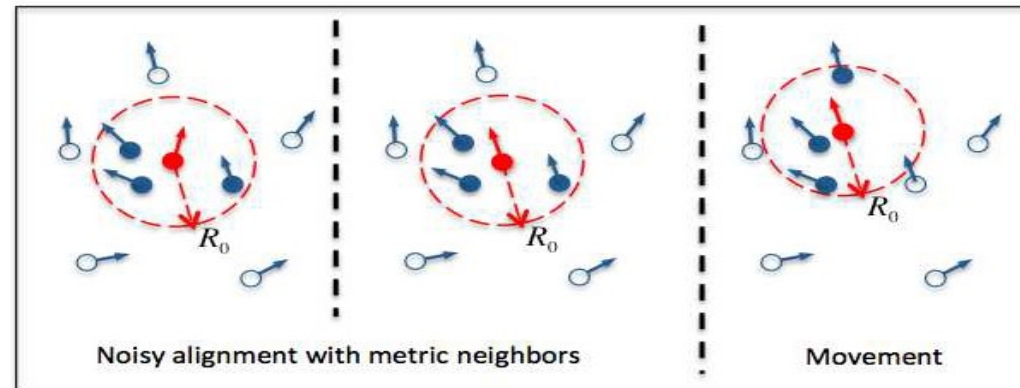
$$\vec{x}_i(t + \Delta t) = \vec{x}_i(t) + \vec{v}_i(t)\Delta t \quad (1)$$

$$\vec{\theta}_i(t + \Delta t) = \text{Arg} [\sum n_{ij}(t)s_j(t)] + \eta \xi_i(t) \quad (2)$$

$\xi_i(t)$ is a zero average, delta correlated scalar noise.

$$\langle \xi_i(t) \rangle = 0, \quad \langle \xi_i(t), \xi_j(k) \rangle \sim \delta_{tk} \delta_{ij}$$

A



Vicsek Model

$$\vec{x}_i(t + \Delta t) = \vec{x}_i(t) + \vec{v}_i(t)\Delta t \quad (1)$$

$$\vec{\theta}_i(t + \Delta t) = \text{Arg} [\sum n_{ij}(t)s_j(t)] + \eta \xi_i(t) \quad (2)$$

$\xi_i(t)$ is a zero average, delta correlated scalar noise.

$$\langle \xi_i(t) \rangle = 0, \quad \langle \xi_i(t), \xi_j(k) \rangle \sim \delta_{tk} \delta_{ij}$$

$$s = (\cos \theta, \sin \theta)$$

In Eq. (2), the function Arg returns the angle defining the orientation of the average vector $\sum_j n_{ij}^t s_j^t$, and n_{ij}^t is the connectivity matrix,

$$n_{ij}^t = \begin{cases} 1 & \text{if } |\mathbf{r}_i^t - \mathbf{r}_j^t| < R_0 \\ 0 & \text{if } |\mathbf{r}_i^t - \mathbf{r}_j^t| > R_0 \end{cases} \quad (4)$$

Vicsek Model

$$\vec{x}_i(t + \Delta t) = \vec{x}_i(t) + \vec{v}_i(t)\Delta t$$

$$\vec{\theta}_i(t + \Delta t) = \text{Arg} [\sum n_{ij}(t)s_j(t)] + \eta\xi_i(t)$$

- The Vicsek model is defined in terms of N point-like, off lattice particles, with positions $\vec{x}_i$.
- Velocities, having fixed magnitude v_o , denoted by $\vec{v}_i \quad i = 1; \dots; N$.
- t and $t + \Delta t$ successive time steps.
- In two dimensions, the direction of motion of the i -th particle can be described via a single angle, θ_i , so $\vec{v}_i = (v_o \cos \theta_i, v_o \sin \theta_i)$
(T. Vicsek *et al.*, PRL, 1995)

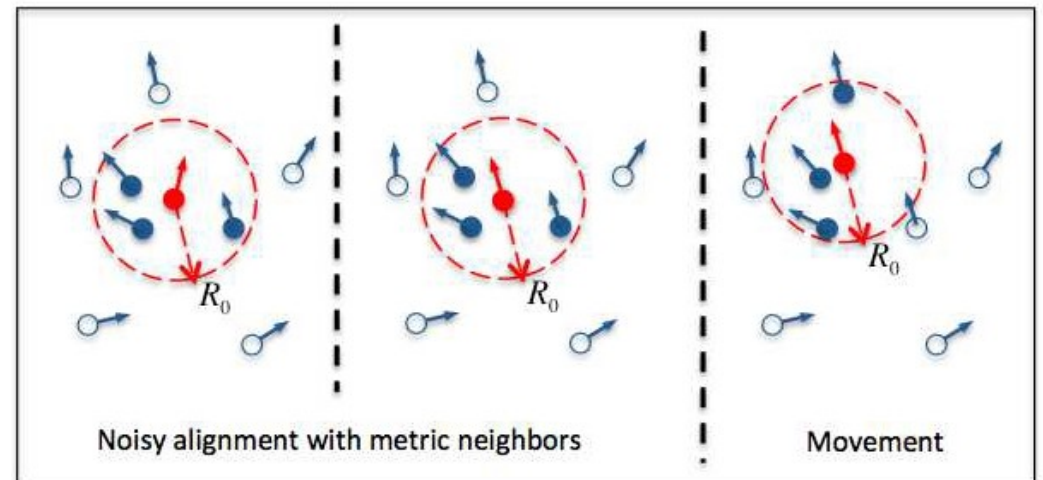
Vicsek Model

$$\vec{x}_i(t + \Delta t) = \vec{x}_i(t) + \vec{v}_i(t)\Delta t$$

$$\vec{\theta}_i(t + \Delta t) = \text{Arg} [\sum n_{ij}(t)s_j(t)] + \eta\xi_i(t)$$

- standard periodic boundary conditions in all directions were applied.
- For suitability take $R_0 = 1, \Delta t = 1$

A



Vicsek Model

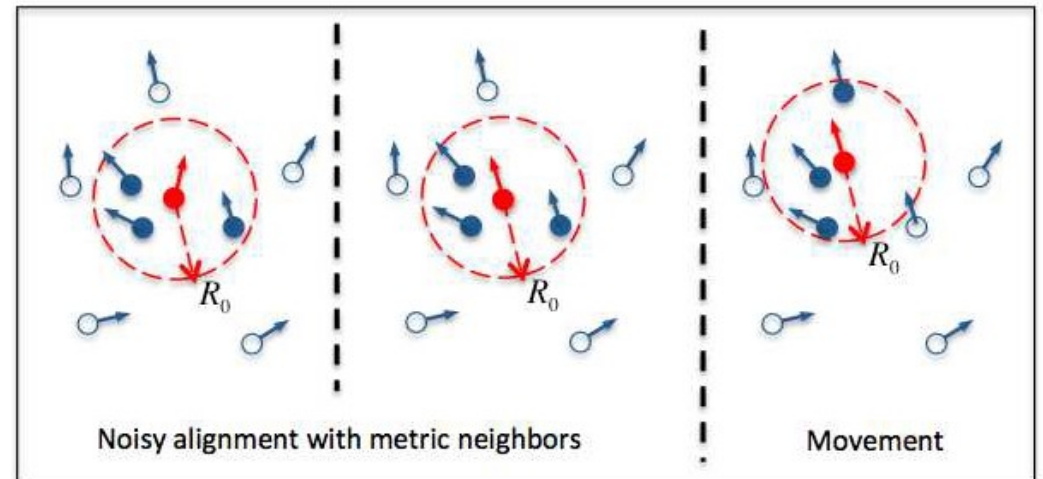
$$\vec{x}_i(t + \Delta t) = \vec{x}_i(t) + \vec{v}_i(t)\Delta t$$

$$\vec{\theta}_i(t + \Delta t) = \text{Arg} [\sum n_{ij}(t)s_j(t)] + \eta\xi_i(t)$$

- standard periodic boundary conditions in all directions were applied.
- For suitability take $t = R_0 = 1$

- *Connectivity matrix is not static*
- *Number of particles are not changing*

A



Vicsek Model

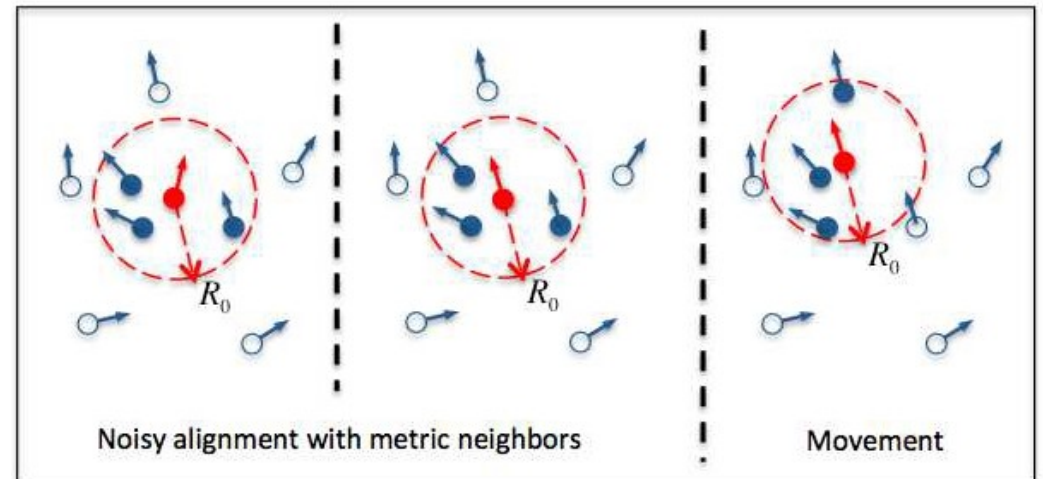
$$\vec{x}_i(t + \Delta t) = \vec{x}_i(t) + \vec{v}_i(t)\Delta t$$

$$\vec{\theta}_i(t + \Delta t) = \text{Arg} [\sum n_{ij}(t)s_j(t)] + \eta\xi_i(t)$$

- standard periodic boundary conditions in all directions were applied.
- For suitability take $R_0 = 1, \Delta t = 1$

- The particle in red aligns imperfectly with neighbours inside the metric range R_0 (dark blue) and then moves along its heading direction.
- Note that neighbours may change as a consequence of movement.
- In the cartoon only the red particle adjusts its position and heading, but in the full algorithm all particles update synchronously.

A



Algorithm 3 Vicsek Model with Velocity Order Parameter

- 1: Initialize N, L, v_0, r, η, T
- 2: Initialize positions $\mathbf{x}_i$ uniformly in $[0, L]$
- 3: Initialize directions $\theta_i \in [0, 2\pi]$
- 4: **for** $t = 1$ to T **do**
- 5: **for** each particle i **do**
- 6: Find neighbors:

$$\mathcal{N}_i = \{j \mid \|\mathbf{x}_j - \mathbf{x}_i\| \leq r\}$$

- 7: Compute average direction:

$$\bar{\theta}_i = \text{atan2} \left(\sum_{j \in \mathcal{N}_i} \sin \theta_j, \sum_{j \in \mathcal{N}_i} \cos \theta_j \right)$$

- 8: Update direction with noise:

$$\theta_i \leftarrow \bar{\theta}_i + \eta \xi_i$$

- 9: **end for**
- 10: **for** each particle i **do**
- 11: Velocity:

$$\mathbf{v}_i = v_0(\cos \theta_i, \sin \theta_i)$$

- 12: Update position:

$$\mathbf{x}_i \leftarrow \mathbf{x}_i + \mathbf{v}_i$$

- 13: Apply periodic boundary:

$$\mathbf{x}_i \leftarrow \mathbf{x}_i \bmod L$$

- 14: **end for**
- 15: Compute normalized average velocity:

$$\mathbf{V}(t) = \frac{1}{N} \sum_{i=1}^N \mathbf{v}_i$$

Vicsek Model

Transition to collective motion and phase separation

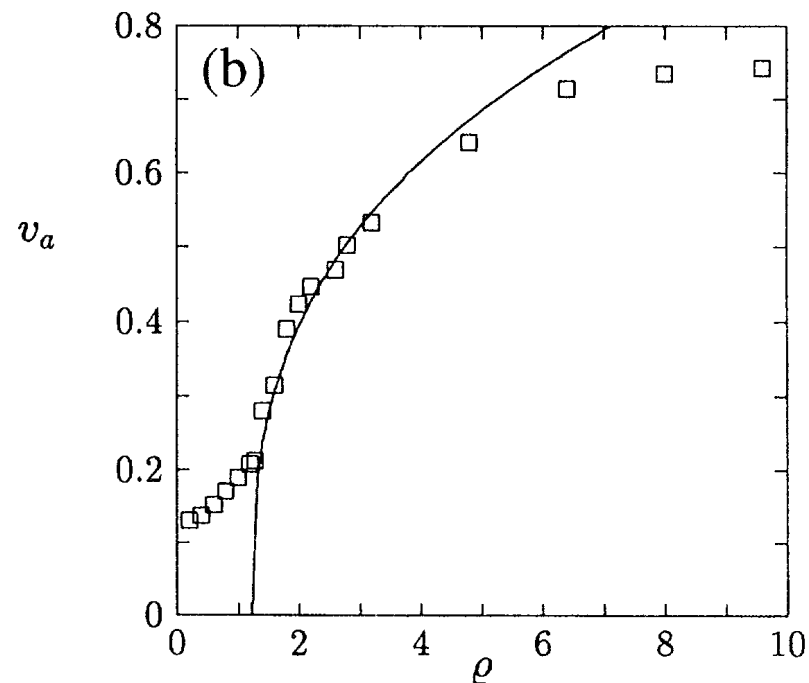
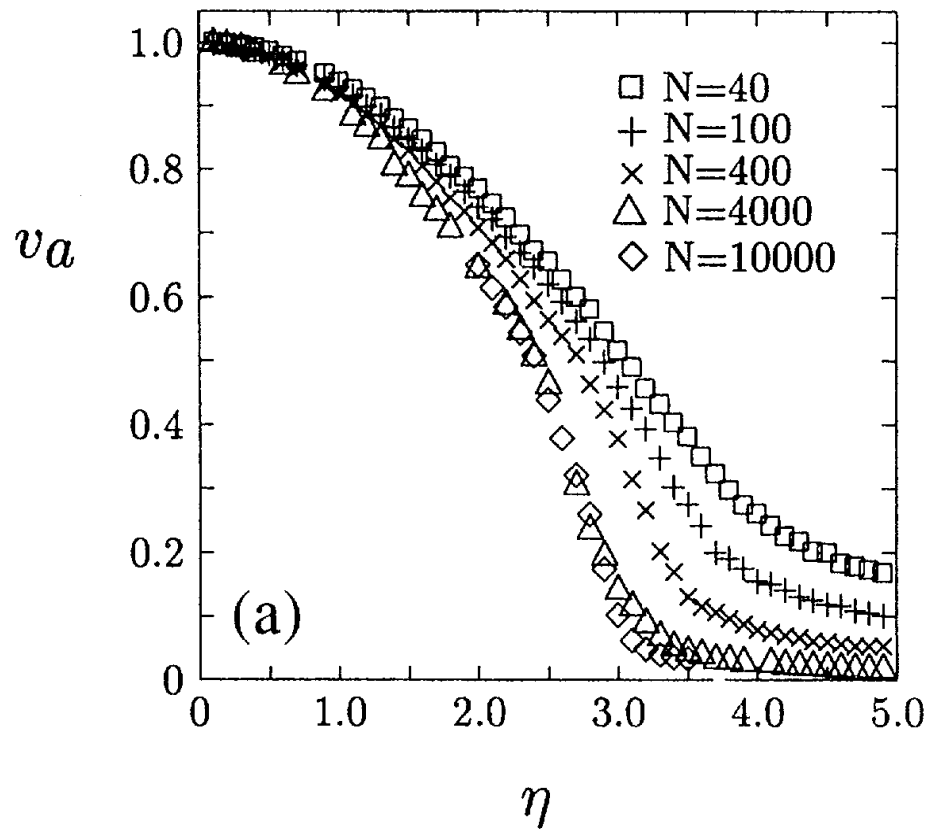
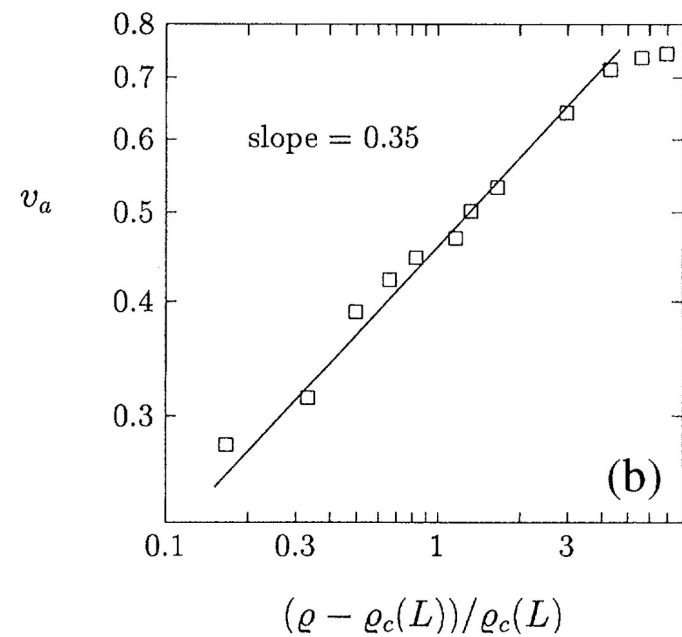
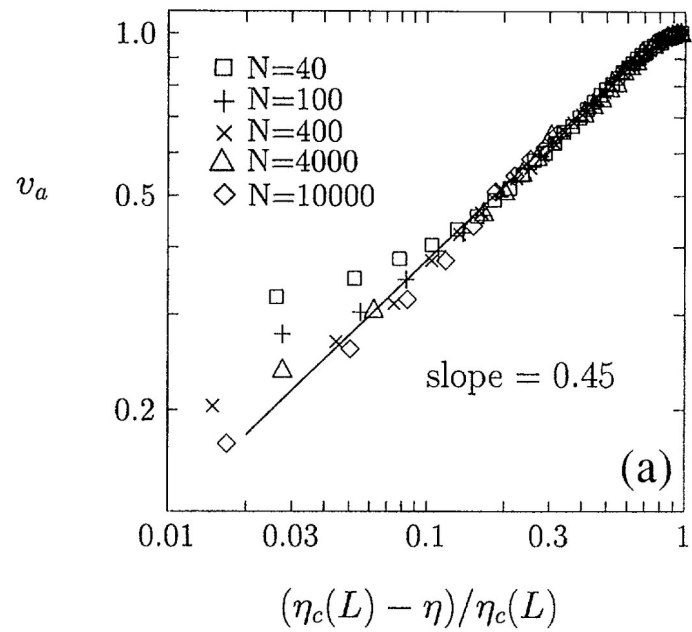
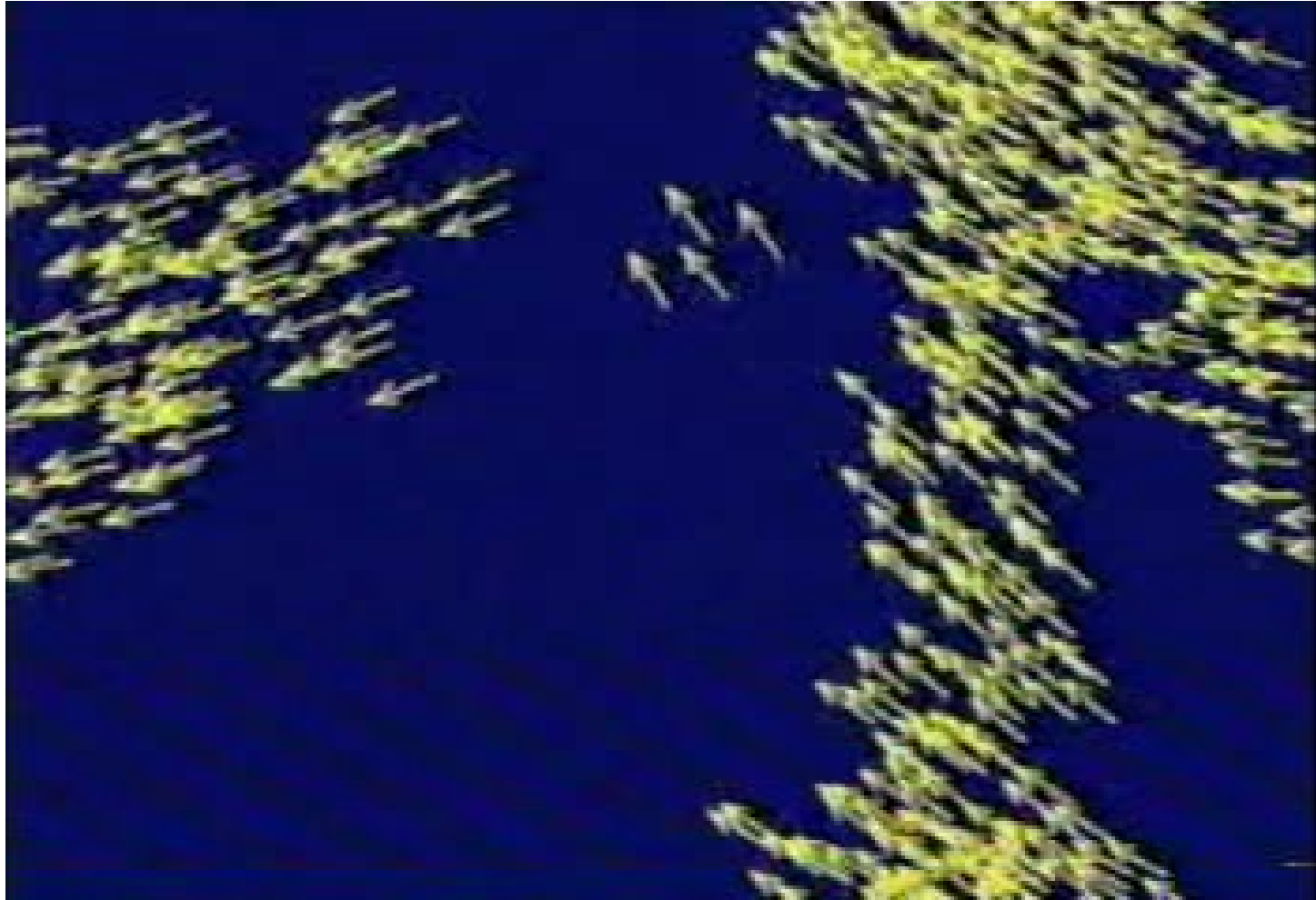


FIG. 2. (a) The absolute value of the average velocity (v_a) versus the noise η in cells of various sizes for a fixed density ρ . The symbols correspond to $\square$: $N = 40$, $L = 3.1$; $+$: $N = 100$, $L = 5$; $\times$: $N = 400$, $L = 10$; $\triangle$: $N = 4000$, $L = 31.6$; $\diamond$: $N = 10000$, $L = 50$. In (b) (for $L = 20$) we show how v_a changes if the noise is kept constant and the density is increased.



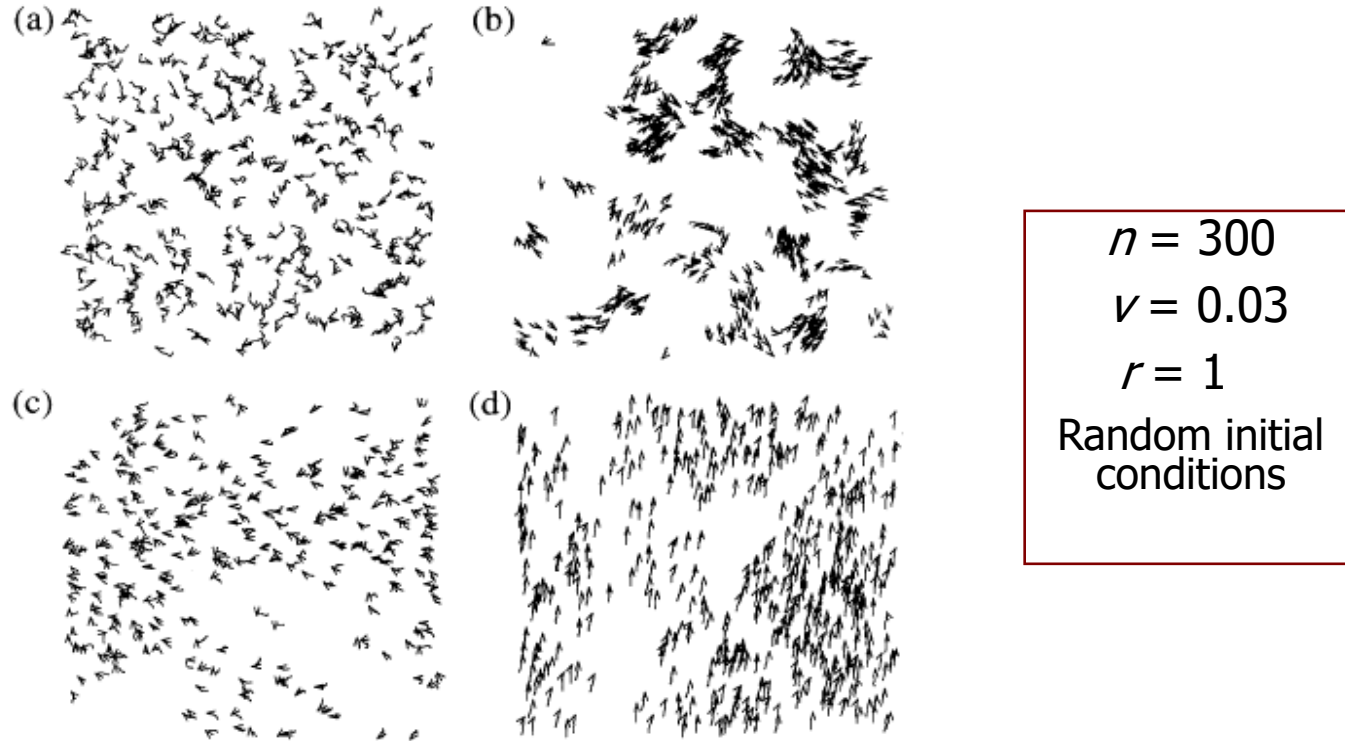
$$v_a \sim [\eta_c(\rho) - \eta]^\beta \quad \text{and} \quad v_a \sim [\rho - \rho_c(\eta)]^\delta,$$

Vicsek Model



Some Phenomena Observed

(Vicsek, *et al.* Physical Review Letters, 1995)



a) $\rho = 6, \epsilon = 1$ high density, large noise

c) 

b) $\rho = 0.48, \epsilon = 0.05$ small density, small noise

d) $\rho = 12, \epsilon = 0.05$ higher density, small noise

Vicsek Model

$$\vec{x}_i(t + \Delta t) = \vec{x}_i(t) + \vec{v}_i(t)\Delta t$$

$$\vec{\theta}_i(t + \Delta t) = \text{Arg} [\sum n_{ij}(t)s_j(t)] + \eta\xi_i(t)$$

Finite polar order parameter (or center of mass velocity)

$$\varphi(t) = \frac{1}{N} \sum_{i=1}^N \mathbf{s}_i^t$$

Stationary time average $\phi = \langle \Phi(t) \rangle_t$

Variant of Vicsek Model

Shear term : an imposed linear velocity profile along the x direction,

$$\vec{x}_i(t + \Delta t) = \vec{x}_i(t) + (\vec{v}_i(t) + \dot{\gamma} y \hat{e}_x) \Delta t$$

$$\theta_i(t + \Delta t) = \text{Arg} \left[\sum_{j \sim i} \exp(i\theta_j(t)) \right] + \eta \psi_i(t),$$

The Vicsek model is extremely simple and particularly well suited for numerical studies, as Eqs. (1)-(2) can be easily implemented on a computer.

However, it should be noted that a straightforward implementation of the metric neighbouring condition would require testing the distance of all $i - j$ couples, an operation scaling with system size as order N^2 . This approach would quickly become unmanageable as the number of SPPs N grows, making practically impossible to run simulations with more than a few thousands particles.

The Vicsek model is extremely simple and particularly well suited for numerical studies, as Eqs. (1)-(2) can be easily implemented on a computer.

However, it should be noted that a straightforward implementation of the metric neighbouring condition would require testing the distance of all $i - j$ couples, an operation scaling with system size as order N^2 . This approach would quickly become unmanageable as the number of SPPs N grows, making practically impossible to run simulations with more than a few thousands particles.

There is of course a way around this problem, based on techniques originally developed for the study of molecular dynamics. The idea is rather simple, even if its algorithmic interpretation may not be so straightforward.

One should ideally divide the system volume L^d in boxes of linear size R_0 (remember that one can always rescale space so that $R_0 = 1$), assigning at each timestep each particle to a given box.

Once this is done, it is clear that for any given particle i , all other particles laying outside the box containing i and its next neighbouring boxes cannot be closer than R_0 . In $d = 2$ one has to only look into 9 boxes (the general formula in spatial dimension d of course gives 3^d boxes).

- Cellular Automata (Hiroki Samaya)
- Flocking of birds analysis

<https://www.complexity-explorables.org/explorables/horde-of-the-flies/>

https://web.mit.edu/8.334/www/grades/projects/projects10/Hernandez-Lopez-Rogelio/dynamics_2.html

https://visualize-it.github.io/collective_behaviour/simulation.html